

2012

**HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11–16, show relevant mathematical reasoning and/or calculations

Total marks – 100

Section I Pages 2–8

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II Pages 9–19

90 marks

- Attempt Questions 11–16
- Allow about 2 hours and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1** Let $z = 5 - i$ and $w = 2 + 3i$.

What is the value of $2z + \bar{w}$?

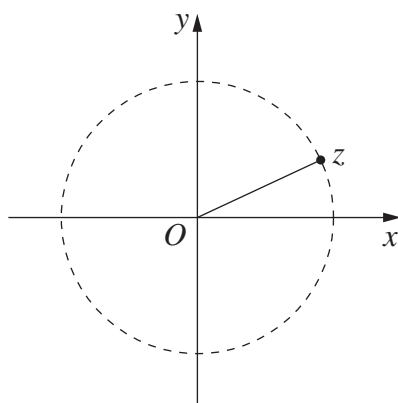
- (A) $12 + i$
- (B) $12 + 2i$
- (C) $12 - 4i$
- (D) $12 - 5i$

- 2** The equation $x^3 - y^3 + 3xy + 1 = 0$ defines y implicitly as a function of x .

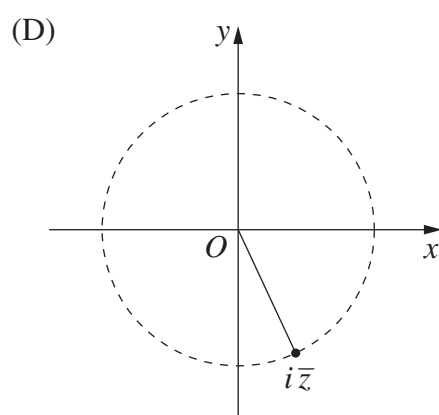
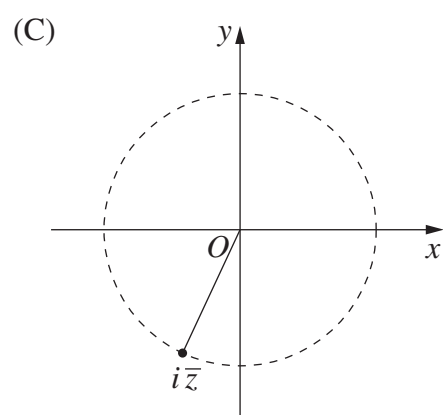
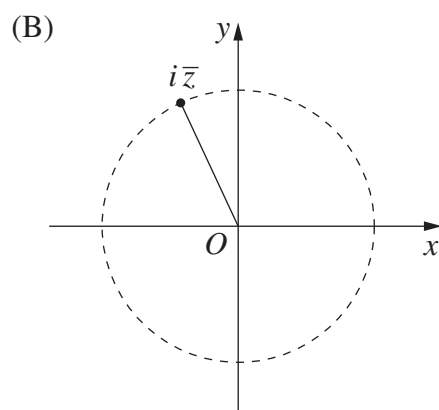
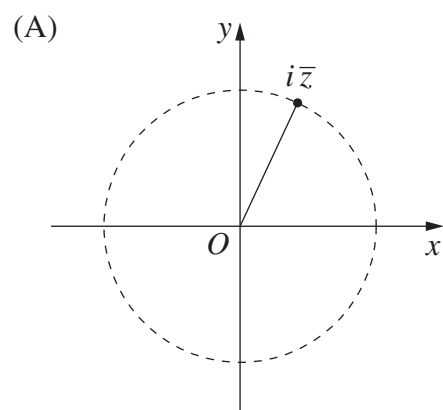
What is the value of $\frac{dy}{dx}$ at the point $(1, 2)$?

- (A) $\frac{1}{3}$
- (B) $\frac{1}{2}$
- (C) $\frac{3}{4}$
- (D) 1

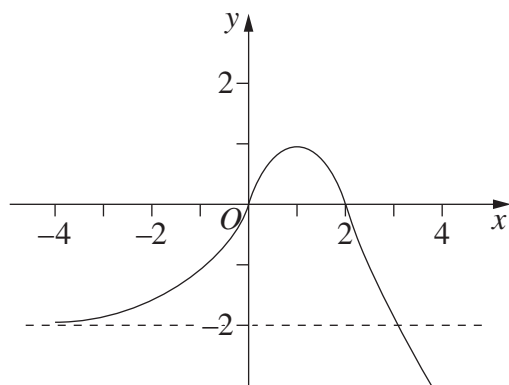
- 3 The complex number z is shown on the Argand diagram below.



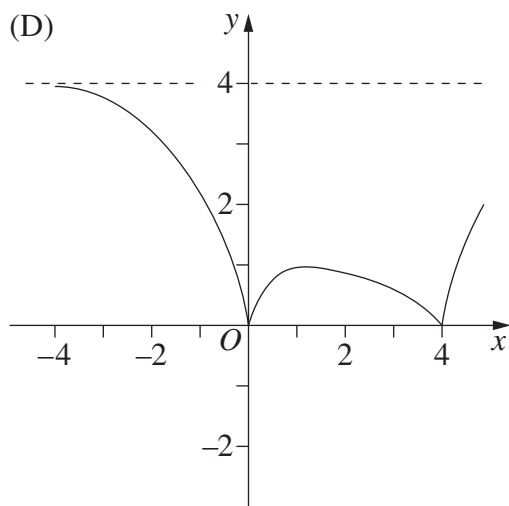
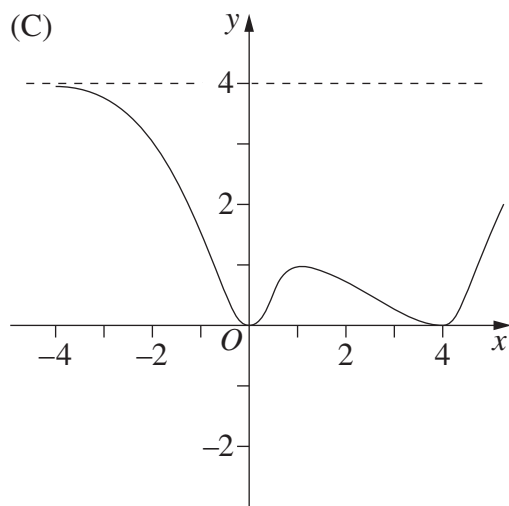
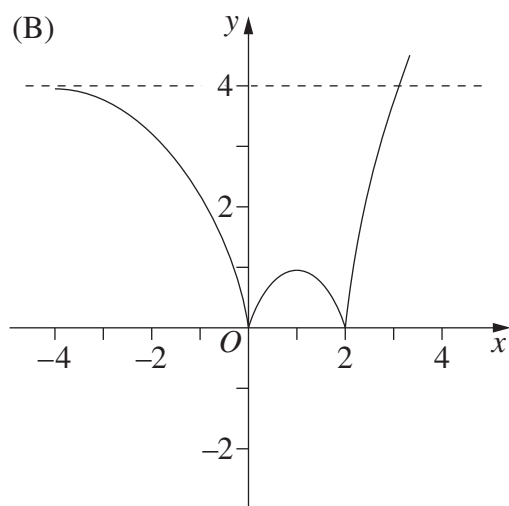
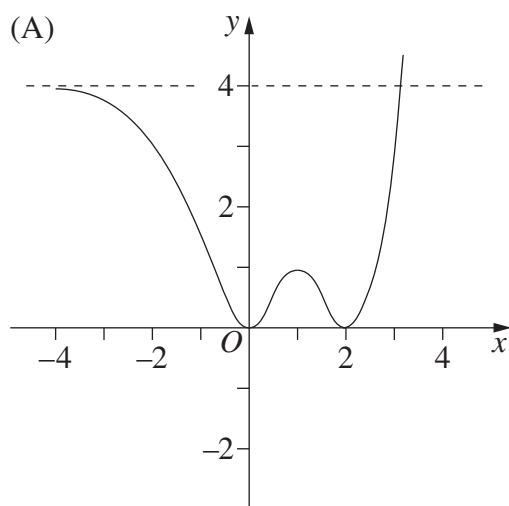
Which of the following best represents $i\bar{z}$?



- 4 The graph $y = f(x)$ is shown below.



Which of the following graphs best represents $y = [f(x)]^2$?



- 5 The equation $2x^3 - 3x^2 - 5x - 1 = 0$ has roots α , β and γ .

What is the value of $\frac{1}{\alpha^3\beta^3\gamma^3}$?

(A) $\frac{1}{8}$

(B) $-\frac{1}{8}$

(C) 8

(D) -8

- 6 What is the eccentricity of the hyperbola $\frac{x^2}{6} - \frac{y^2}{4} = 1$?

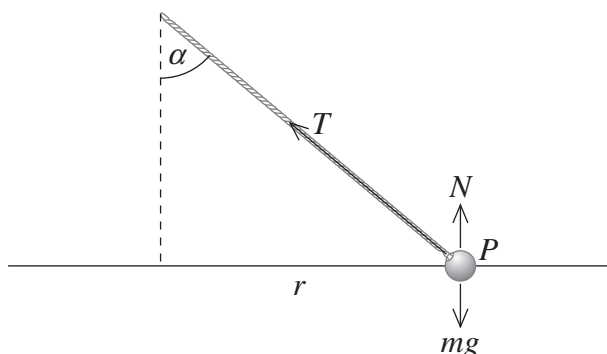
(A) $\frac{\sqrt{10}}{2}$

(B) $\frac{\sqrt{15}}{3}$

(C) $\frac{\sqrt{3}}{3}$

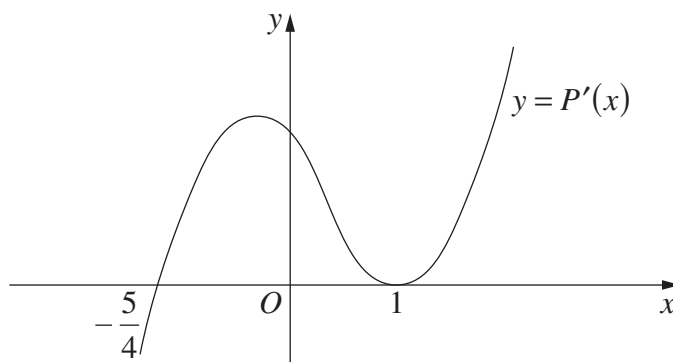
(D) $\frac{\sqrt{13}}{3}$

- 7 A particle P of mass m attached to a string is rotating in a circle of radius r on a smooth horizontal surface. The particle is moving with constant angular velocity ω . The string makes an angle α with the vertical. The forces acting on P are the tension T in the string, a reaction force N normal to the surface and the gravitational force mg .



Which of the following is the correct resolution of the forces on P in the vertical and horizontal directions?

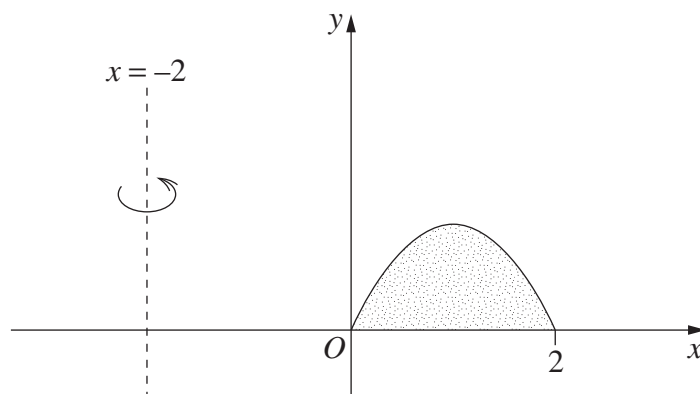
- (A) $T \cos \alpha + N = mg$ and $T \sin \alpha = mr\omega^2$
 (B) $T \cos \alpha - N = mg$ and $T \sin \alpha = mr\omega^2$
 (C) $T \sin \alpha + N = mg$ and $T \cos \alpha = mr\omega^2$
 (D) $T \sin \alpha - N = mg$ and $T \cos \alpha = mr\omega^2$
- 8 The following diagram shows the graph $y = P'(x)$, the derivative of a polynomial $P(x)$.



Which of the following expressions could be $P(x)$?

- (A) $(x - 2)(x - 1)^3$
 (B) $(x + 2)(x - 1)^3$
 (C) $(x - 2)(x + 1)^3$
 (D) $(x + 2)(x + 1)^3$

- 9 The diagram shows the graph $y = x(2 - x)$ for $0 \leq x \leq 2$. The region bounded by the graph and the x -axis is rotated about the line $x = -2$ to form a solid.



Which integral represents the volume of the solid?

- (A) $2\pi \int_0^2 x(2 - x)^2 dx$
- (B) $2\pi \int_0^2 x^2(2 - x) dx$
- (C) $2\pi \int_0^2 x(2 - x)(2 + x) dx$
- (D) $2\pi \int_0^2 x(2 - x)(x - 2) dx$

- 10** Without evaluating the integrals, which one of the following integrals is greater than zero?

(A) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{x}{2 + \cos x} dx$

(B) $\int_{-\pi}^{\pi} x^3 \sin x dx$

(C) $\int_{-1}^1 (e^{-x^2} - 1) dx$

(D) $\int_{-2}^2 \tan^{-1}(x^3) dx$

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

(a) Express $\frac{2\sqrt{5} + i}{\sqrt{5} - i}$ in the form $x + iy$, where x and y are real. **2**

(b) Shade the region on the Argand diagram where the two inequalities **2**

$$|z + 2| \geq 2 \quad \text{and} \quad |z - i| \leq 1$$

both hold.

(c) By completing the square, find $\int \frac{dx}{x^2 + 4x + 5}$. **2**

(d) (i) Write $z = \sqrt{3} - i$ in modulus–argument form. **2**

(ii) Hence express z^9 in the form $x + iy$, where x and y are real. **1**

(e) Evaluate $\int_0^1 \frac{e^{2x}}{e^{2x} + 1} dx$. **3**

(f) Sketch the following graphs, showing the x - and y -intercepts.

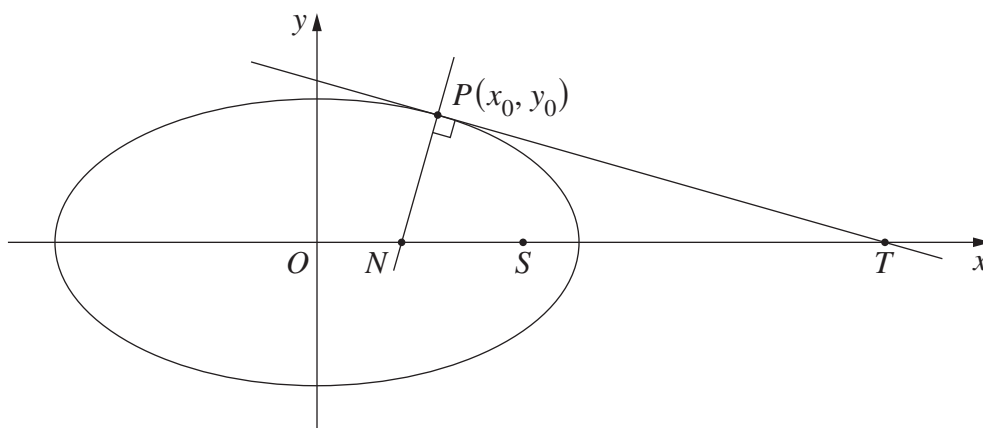
(i) $y = |x| - 1$ **1**

(ii) $y = x(|x| - 1)$ **2**

Question 12 (15 marks) Use a SEPARATE writing booklet.

(a) Using the substitution $t = \tan \frac{\theta}{2}$, or otherwise, find $\int \frac{d\theta}{1 - \cos \theta}$. **3**

- (b) The diagram shows the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with $a > b$. The ellipse has focus S and eccentricity e . The tangent to the ellipse at $P(x_0, y_0)$ meets the x -axis at T . The normal at P meets the x -axis at N .



- (i) Show that the tangent to the ellipse at P is given by the equation **2**

$$y - y_0 = -\frac{b^2 x_0}{a^2 y_0} (x - x_0).$$

- (ii) Show that the x -coordinate of N is $x_0 e^2$. **2**
- (iii) Show that $ON \times OT = OS^2$. **2**

Question 12 continues on page 11

Question 12 (continued)

- (c) For every integer $n \geq 0$ let

3

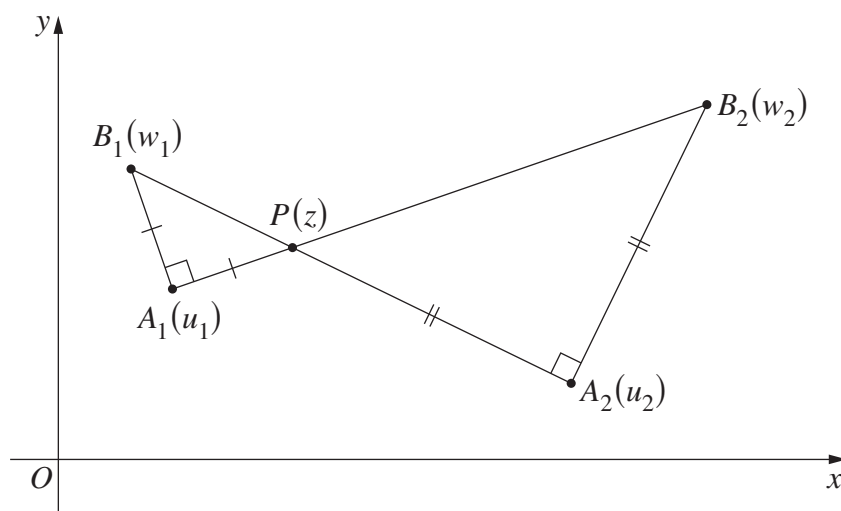
$$I_n = \int_1^{e^2} (\log_e x)^n dx.$$

Show that for $n \geq 1$

$$I_n = e^2 2^n - n I_{n-1}.$$

- (d) On the Argand diagram the points A_1 and A_2 correspond to the distinct complex numbers u_1 and u_2 respectively. Let P be a point corresponding to a third complex number z .

Points B_1 and B_2 are positioned so that $\triangle A_1 P B_1$ and $\triangle A_2 B_2 P$, labelled in an anti-clockwise direction, are right-angled and isosceles with right angles at A_1 and A_2 , respectively. The complex numbers w_1 and w_2 correspond to B_1 and B_2 , respectively.



- (i) Explain why $w_1 = u_1 + i(z - u_1)$. 1
- (ii) Find the locus of the midpoint of $B_1 B_2$ as P varies. 2

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

- (a) An object on the surface of a liquid is released at time $t = 0$ and immediately sinks. Let x be its displacement in metres in a downward direction from the surface at time t seconds.

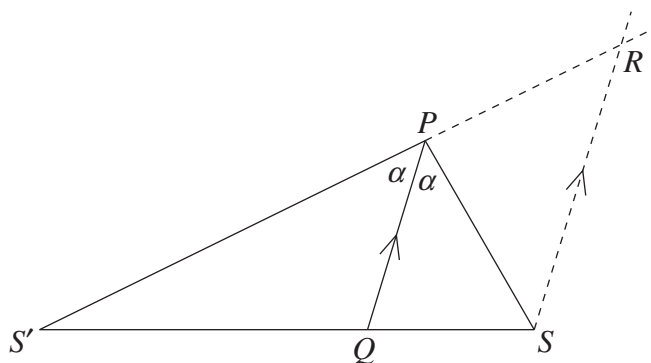
The equation of motion is given by

$$\frac{dv}{dt} = 10 - \frac{v^2}{40},$$

where v is the velocity of the object.

- (i) Show that $v = \frac{20(e^t - 1)}{e^t + 1}$. 4
- (ii) Use $\frac{dv}{dt} = v \frac{dv}{dx}$ to show that $x = 20 \log_e \left(\frac{400}{400 - v^2} \right)$. 2
- (iii) How far does the object sink in the first 4 seconds? 2

- (b) The diagram shows $\triangle S'SP$. The point Q is on $S'S$ so that PQ bisects $\angle S'PS$. The point R is on $S'P$ produced so that $PQ \parallel RS$.

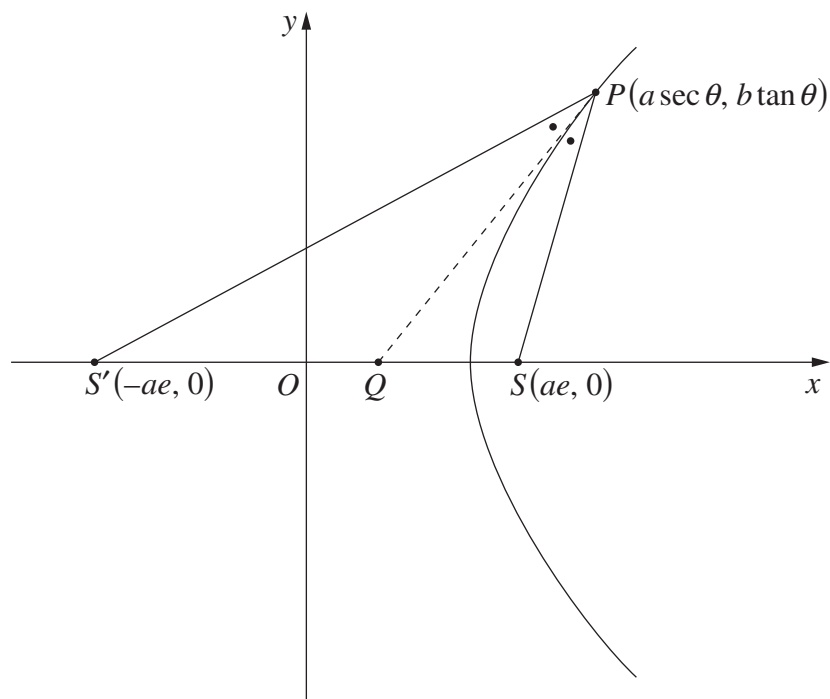


- (i) Show that $PS = PR$. 1
- (ii) Show that $\frac{PS}{QS} = \frac{PS'}{QS'}$. 2

Question 13 continues on page 13

Question 13 (continued)

- (c) Let P be a point on the hyperbola given parametrically by $x = a \sec \theta$ and $y = b \tan \theta$, where a and b are positive. The foci of the hyperbola are $S(ae, 0)$ and $S'(-ae, 0)$ where e is the eccentricity. The point Q is on the x -axis so that PQ bisects $\angle SPS'$.



- (i) Show that $SP = a(e \sec \theta - 1)$. 1
- (ii) It is given that $S'P = a(e \sec \theta + 1)$. Using part (b), or otherwise, show that the x -coordinate of Q is $\frac{a}{\sec \theta}$. 2
- (iii) The slope of the tangent to the hyperbola at P is $\frac{b \sec \theta}{a \tan \theta}$. (Do NOT prove this.) 1

Show that the tangent at P is the line PQ .

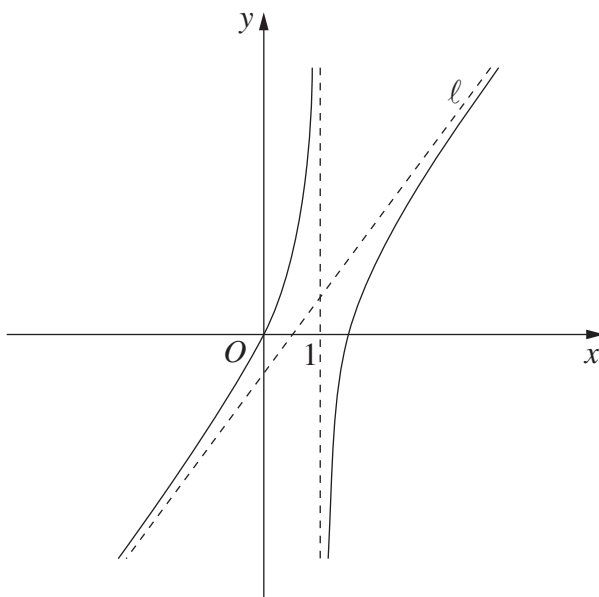
End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

(a) Find $\int \frac{3x^2 + 8}{x(x^2 + 4)} dx$.

3

(b) The diagram shows the graph $y = \frac{x(2x - 3)}{x - 1}$. The line ℓ is an asymptote.



(i) Use the above graph to draw a one-third page sketch of the graph

2

$$y = \frac{x - 1}{x(2x - 3)}$$

indicating all asymptotes and all x - and y -intercepts.

(ii) By writing $\frac{x(2x - 3)}{x - 1}$ in the form $mx + b + \frac{a}{x - 1}$, find the equation of the line ℓ .

2

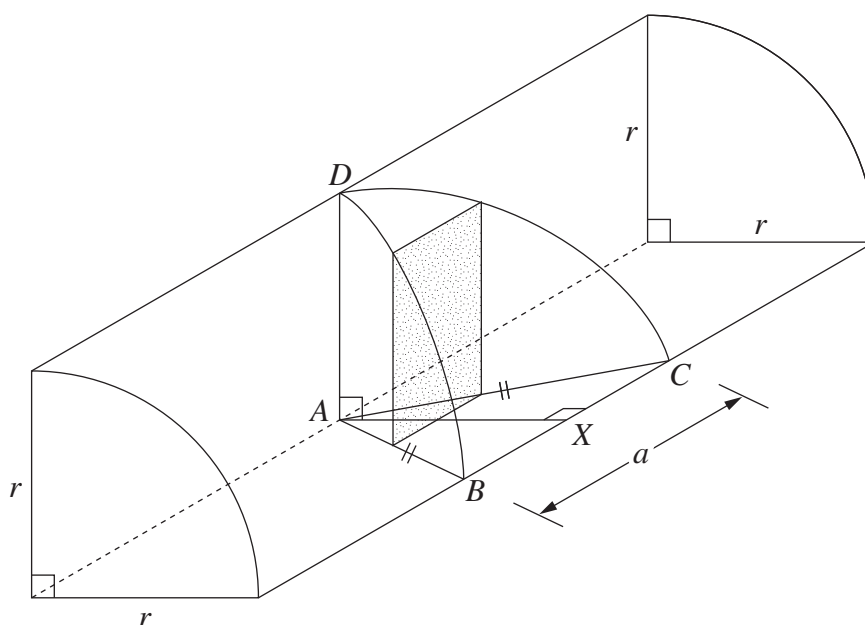
Question 14 continues on page 15

Question 14 (continued)

- (c) The solid $ABCD$ is cut from a quarter cylinder of radius r as shown. Its base is an isosceles triangle ABC with $AB = AC$. The length of BC is a and the midpoint of BC is X .

4

The cross-sections perpendicular to AX are rectangles. A typical cross-section is shown shaded in the diagram.

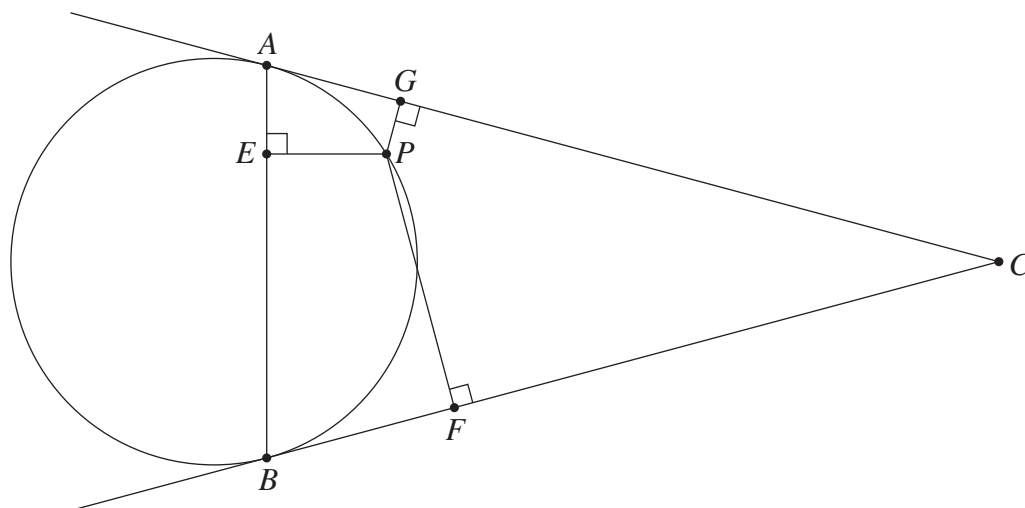


Find the volume of the solid $ABCD$.

Question 14 continues on page 16

Question 14 (continued)

- (d) The diagram shows points A and B on a circle. The tangents to the circle at A and B meet at the point C . The point P is on the circle inside $\triangle ABC$. The point E lies on AB so that $AB \perp EP$. The points F and G lie on BC and AC respectively so that $FP \perp BC$ and $GP \perp AC$.



Copy or trace the diagram into your writing booklet.

- | | |
|--|----------|
| (i) Show that $\triangle APG$ and $\triangle BPE$ are similar. | 2 |
| (ii) Show that $EP^2 = FP \times GP$. | 2 |

End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet.

(a) (i) Prove that $\sqrt{ab} \leq \frac{a+b}{2}$, where $a \geq 0$ and $b \geq 0$. 1

(ii) If $1 \leq x \leq y$, show that $x(y-x+1) \geq y$. 2

(iii) Let n and j be positive integers with $1 \leq j \leq n$. Prove that 2

$$\sqrt{n} \leq \sqrt{j(n-j+1)} \leq \frac{n+1}{2}.$$

(iv) For integers $n \geq 1$, prove that 1

$$(\sqrt{n})^n \leq n! \leq \left(\frac{n+1}{2}\right)^n.$$

(b) Let $P(z) = z^4 - 2kz^3 + 2k^2z^2 - 2kz + 1$, where k is real.

Let $\alpha = x + iy$, where x and y are real.

Suppose that α and $i\alpha$ are zeros of $P(z)$, where $\bar{\alpha} \neq i\alpha$.

(i) Explain why $\bar{\alpha}$ and $-i\bar{\alpha}$ are zeros of $P(z)$. 1

(ii) Show that $P(z) = z^2(z-k)^2 + (kz-1)^2$. 1

(iii) Hence show that if $P(z)$ has a real zero then 2

$$P(z) = (z^2+1)(z+1)^2 \text{ or } P(z) = (z^2+1)(z-1)^2.$$

(iv) Show that all zeros of $P(z)$ have modulus 1. 2

(v) Show that $k = x - y$. 1

(vi) Hence show that $-\sqrt{2} \leq k \leq \sqrt{2}$. 2

Question 16 (15 marks) Use a SEPARATE writing booklet.

- (a) (i) In how many ways can m identical yellow discs and n identical black discs be arranged in a row? **1**
- (ii) In how many ways can 10 identical coins be allocated to 4 different boxes? **1**

- (b) (i) Show that $\tan^{-1}x + \tan^{-1}y = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$ for $|x| < 1$ and $|y| < 1$. **1**
- (ii) Use mathematical induction to prove **3**

$$\sum_{j=1}^n \tan^{-1}\left(\frac{1}{2j^2}\right) = \tan^{-1}\left(\frac{n}{n+1}\right)$$

for all positive integers n .

- (iii) Find $\lim_{n \rightarrow \infty} \sum_{j=1}^n \tan^{-1}\left(\frac{1}{2j^2}\right)$. **1**

Question 16 continues on page 19

Question 16 (continued)

- (c) Let n be an integer where $n > 1$. Integers from 1 to n inclusive are selected randomly one by one with repetition being possible. Let $P(k)$ be the probability that exactly k different integers are selected before one of them is selected for the second time, where $1 \leq k \leq n$.

(i) Explain why $P(k) = \frac{(n-1)!k}{n^k(n-k)!}$. 2

(ii) Suppose $P(k) \geq P(k-1)$. Show that $k^2 - k - n \leq 0$. 2

(iii) Show that if $\sqrt{n + \frac{1}{4}} > k - \frac{1}{2}$ then the integers n and k satisfy 2
 $\sqrt{n} > k - \frac{1}{2}$.

(iv) Hence show that if $4n + 1$ is not a perfect square, then $P(k)$ is greatest when k is the closest integer to $\sqrt{n}$. 2

You may use part (iii) and also that $k^2 - k - n > 0$ if $P(k) < P(k-1)$.

End of paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$



2012 HSC Mathematics Extension 2 Marking Guidelines

Section I

Multiple-choice Answer Key

Question	Answer
1	D
2	D
3	A
4	A
5	C
6	B
7	A
8	B
9	C
10	B

Section II

Question 11 (a)

Criteria	Marks
• Correct solution	2
• Writes $\frac{2\sqrt{5} + i}{\sqrt{5} - i} \times \frac{\sqrt{5} + i}{\sqrt{5} + i}$, or equivalent merit	1

Question 11 (b)

Criteria	Marks
• Correct diagram	2
• Two correct circles or correct region for one of the inequalities, or equivalent merit	1

Question 11 (c)

Criteria	Marks
• Correct solution	2
• Correct completion of square, or equivalent merit	1

Question 11 (d) (i)

Criteria	Marks
• Correct answer	2
• Correct modulus or argument, or equivalent merit	1

Question 11 (d) (ii)

Criteria	Marks
• Correct answer	1

Question 11 (e)

Criteria	Marks
• Correct solution	3
• Correct primitive or correct substitution of limits into primitive of the form $\ln(e^{2x} + 1)$, or equivalent merit	2
• Primitive of the form $\ln(e^{2x} + 1)$, or equivalent merit	1

Question 11 (f) (i)

Criteria	Marks
• Correct graph	1

Question 11 (f) (ii)

Criteria	Marks
• Correct graph	2
• Graph correct for $x > 0$, or equivalent merit	1

Question 12 (a)

Criteria	Marks
• Correct solution	3
• Correct integrand in terms of t , or equivalent merit	2
• Correct $d\theta$ or $\cos \theta$ in terms of t , or equivalent merit	1

Question 12 (b) (i)

Criteria	Marks
• Correct solution	2
• Finds $\frac{dy}{dx}$, or equivalent merit	1

Question 12 (b) (ii)

Criteria	Marks
• Correct solution	2
• Substitutes $y = 0$ into equation of the normal, or equivalent merit	1

Question 12 (b) (iii)

Criteria	Marks
• Correct solution	2
• Finds correct x -coordinate of T , or equivalent merit	1

Question 12 (c)

Criteria	Marks
• Correct solution	3
• Correct integration by parts, or equivalent merit	2
• Attempts a relevant integration by parts, or equivalent merit	1

Question 12 (d) (i)

Criteria	Marks
• Correct explanation	1

Question 12 (d) (ii)

Criteria	Marks
• Correct locus	2
• Correct expression for w_2 , or equivalent merit	1

Question 13 (a) (i)

Criteria	Marks
• Correct solution	4
• Correct expression of e^t in terms of v , or equivalent merit	3
• Correctly integrates $\frac{dt}{dv}$, or equivalent merit	2
• Correct expression for $\frac{dt}{dv}$, or equivalent merit	1

Question 13 (a) (ii)

Criteria	Marks
• Correct solution	2
• Correct expression for $\frac{dx}{dv}$, or equivalent merit	1

Question 13 (a) (iii)

Criteria	Marks
• Correct solution	2
• Correct value of v , or equivalent merit	1

Question 13 (b) (i)

Criteria	Marks
• Correct proof	1

Question 13 (b) (ii)

Criteria	Marks
• Correct proof	2
• Uses properties of similar triangle or intercepts on parallel lines, or equivalent merit	1

Question 13 (c) (i)

Criteria	Marks
• Correct solution	1

Question 13 (c) (ii)

Criteria	Marks
• Correct solution	2
• Obtains a correct equation for the x -coordinate of Q , or equivalent merit	1

Question 13 (c) (iii)

Criteria	Marks
• Correct solution	1

Question 14 (a)

Criteria	Marks
• Correct primitive	3
• Correct partial fraction decomposition of integrand	2
• Attempts to use partial fractions, or equivalent merit	1

Question 14 (b) (i)

Criteria	Marks
• Correct graph	2
• Correct asymptotes, or equivalent merit	1

Question 14 (b) (ii)

Criteria	Marks
• Correct solution	2
• Correct m or b	1

Question 14 (c)

Criteria	Marks
• Correct solution	4
• Correct primitive, or equivalent merit	3
• Correct definite integral for volume, or equivalent merit	2
• Correct width or height or cross-section, or equivalent merit	1

Question 14 (d) (i)

Criteria	Marks
• Correct proof	2
• Finds a pair of corresponding equal angles with justification, or equivalent merit	1

Question 14 (d) (ii)

Criteria	Marks
• Correct solution	2
• Finds a second pair of relevant similar triangles, or a relevant ratio	1

Question 15 (a) (i)

Criteria	Marks
• Correct proof	1

Question 15 (a) (ii)

Criteria	Marks
• Correct proof	2
• Makes some progress	1

Question 15 (a) (iii)

Criteria	Marks
• Correct proof	2
• Proves one of the two inequalities	1

Question 15 (a) (iv)

Criteria	Marks
• Correct proof	1

Question 15 (b) (i)

Criteria	Marks
• Correct explanation	1

Question 15 (b) (ii)

Criteria	Marks
• Correct solution	1

Question 15 (b) (iii)

Criteria	Marks
• Correct solution	2
• Recognises that both terms in part (ii) are zero, or equivalent merit	1

Question 15 (b) (iv)

Criteria	Marks
• Correct solution	2
• Makes some progress	1

Question 15 (b) (v)

Criteria	Marks
• Correct proof	1

Question 15 (b) (vi)

Criteria	Marks
• Correct solution	2
• Attempts to use discriminant or perpendicular distance formula, or equivalent merit	1

Question 16 (a) (i)

Criteria	Marks
• Correct answer	1

Question 16 (a) (ii)

Criteria	Marks
• Correct answer	1

Question 16 (b) (i)

Criteria	Marks
• Correct solution	1

Question 16 (b) (ii)

Criteria	Marks
• Complete induction proof	3
• Makes some progress in the induction step, or equivalent merit	2
• Proves case $n = 1$, or equivalent merit	1

Question 16 (b) (iii)

Criteria	Marks
• Correct answer	1

Question 16 (c) (i)

Criteria	Marks
• Correct explanation	2
• Partial explanation	1

Question 16 (c) (ii)

Criteria	Marks
• Correct solution	2
• Attempts to solve $\frac{(n-1)!k}{n^k(n-k)!} \geq \frac{(n-1)!(k-1)}{n^{k-1}(n-k+1)!}$	1

Question 16 (c) (iii)

Criteria	Marks
• Correct solution	2
• Obtains $n > k^2 - k$, or equivalent merit	1

Question 16 (c) (iv)

Criteria	Marks
• Correct solution	2
• Finds that k is the largest integer less than or equal to $\frac{1 + \sqrt{1 + 4n}}{2}$, or equivalent merit	1

Mathematics Extension 2

2012 HSC Examination Mapping Grid

Section I

Question	Marks	Content	Syllabus outcomes
1	1	2.1	E3
2	1	1.8	E6
3	1	2.2	E3
4	1	1.6	E6
5	1	7.5	E4
6	1	3.2	E4
7	1	6.3	E5
8	1	7.2, 8.0	HE4, E4
9	1	5.1	E7
10	1	4.1	E8

Section II

Question	Marks	Content	Syllabus outcomes
11 (a)	2	2.1	E3
11 (b)	2	2.5	E3
11 (c)	2	4.1	E8
11 (d) (i)	2	2.2	E3
11 (d) (ii)	1	2.4	E3
11 (e)	3	4.1	E8
11 (f) (i)	1	1.3	E6
11 (f) (ii)	2	1.4	E6
12 (a)	3	4.1	E8
12 (b) (i)	2	3.1	E3, E4
12 (b) (ii)	2	3.1	E4
12 (b) (iii)	2	3.1	E3, E4
12 (c)	3	4.1	E8
12 (d) (i)	1	2.2	E3
12 (d) (ii)	2	2.1, 2.2	E3
13 (a) (i)	4	6.2.3	E5
13 (a) (ii)	2	6.2.3	E5
13 (a) (iii)	2	6.2.3	E5
13 (b) (i)	1	8.0	H5, E2
13 (b) (ii)	2	8.0	H5, E2
13 (c) (i)	1	3.2	E3
13 (c) (ii)	2	3.2	E4
13 (c) (iii)	1	3.2	E4

Question	Marks	Content	Syllabus outcomes
14 (a)	3	4.1, 7.6	E8
14 (b) (i)	2	1.5	E6
14 (b) (ii)	2	1.5, 7.6	E4
14 (c)	4	5.1	E7
14 (d) (i)	2	8.1	PE3, E2, E9
14 (d) (ii)	2	8.1	PE3, E2, E9
15 (a) (i)	1	8.3	PE6, E2
15 (a) (ii)	2	8.3	PE6, E2
15 (a) (iii)	2	8.3	PE6, E9
15 (a) (iv)	1	8.3	PE6, E9
15 (b) (i)	1	7.4	E3, E4
15 (b) (ii)	1	7.4, 8.0	PE3, E2
15 (b) (iii)	2	7.4	E3, E4
15 (b) (iv)	2	7.5	E3
15 (b) (v)	1	7.5	E3
15 (b) (vi)	2	2.5	E2, E9
16 (a) (i)	1	8.0	PE3, E2, E9
16 (a) (ii)	1	8.0	PE3, E2, E9
16 (b) (i)	1	8.0	HE4, E2
16 (b) (ii)	3	8.2	HE2, E2, E9
16 (b) (iii)	1	8.0	HE7, E2
16 (c) (i)	2	8.0	PE3, E9
16 (c) (ii)	2	8.3	PE6, E2
16 (c) (iii)	2	8.3	PE6, E9
16 (c) (iv)	2	8.0, 8.3	PE6, E9